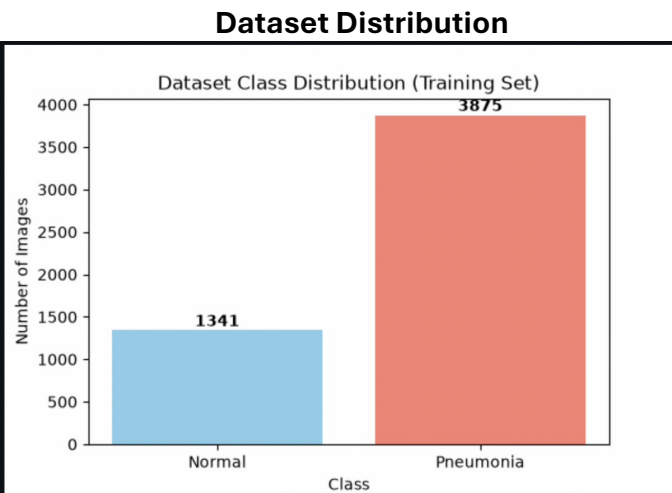


Pneumonia Detection Using Deep Learning Convolutional Neural Networks (CNN)

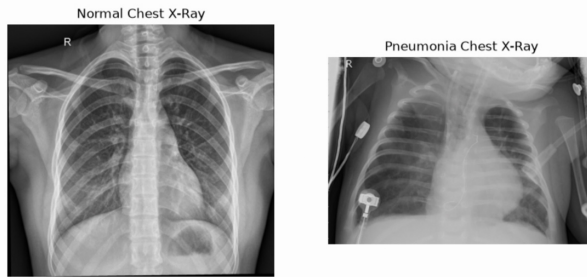
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1. Introduction

In the medical field, early and accurate diagnosis of respiratory conditions like pneumonia is critical for effective patient treatment. Chest X-rays are standard diagnostic tools, but manual analysis can be time-consuming and prone to human variability. Automated screening systems powered by Deep Learning can assist medical professionals by rapidly analyzing chest radiographs to detect signs of infection. The core function of this project is to implement an end-to-end Convolutional Neural Network (CNN) pipeline that classifies chest X-ray images into two distinct categories: **Normal** or **Pneumonia**. By automating this feature extraction and classification process, the model learns spatial hierarchies and visual patterns (such as lung opacities and consolidations) directly from pixel data.



Sample Examples



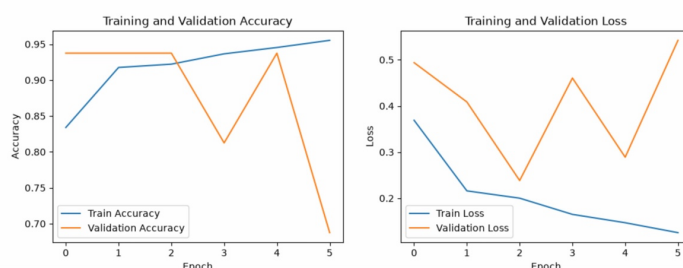
2. Problem Statement

The primary objective of this project is to build, train, and evaluate a robust deep learning classifier capable of accurately distinguishing between normal chest radiographs and those exhibiting clinical signs of pneumonia.

- **Input:** Chest X-ray images resized to standardized dimensions ($150 \times 150 \times 3$).
- **Output / Label:** Binary classification (0 for Normal, 1 for Pneumonia).
- **Dataset:** Chest X-ray image dataset split into structured training, validation, and testing sets comprising diverse patient scans with varying contrast and resolution.

The problem requires constructing a multi-stage workflow encompassing exploratory data analysis (EDA), image preprocessing/tensor verification, custom CNN architecture design, performance evaluation via confusion matrices, and a functional single-image inference script.

Model Training Progress



3. Methodology & Pipeline Implementation

The project follows a rigorous, structured machine learning pipeline divided into modular components:

1. **Exploratory Data Analysis (01_edu.ipynb):** Examined dataset distributions, class balances, and visualized sample X-ray scans to understand pixel intensity variations and structural layouts.
2. **Data Preprocessing (02_preprocessing.ipynb):** Verified image tensor shapes, applied pixel rescaling (normalizing values between 0.0 and 1.0), and configured generator pipelines for batch processing.
3. **Model Training (03_model_training.ipynb):** Designed a custom sequential CNN architecture featuring alternating convolutional and max-pooling layers,

compiled with binary cross-entropy loss and optimized via Adam. Trained weights were securely serialized and stored in the models/ directory (pneumonia_cnn_model.keras).

4. **Evaluation (04_evaluation.ipynb):** Assessed model generalization on unseen test data, computing accuracy, precision, recall, and generating a professional confusion matrix saved in the outputs/ folder.
5. **Inference Engine (05_predict_single_image.ipynb):** Developed a custom prediction script incorporating threshold management to analyze individual X-ray paths from text input and output real-time confidence scores.

4. Results and Discussion

After training and evaluating the custom CNN model on the test dataset, the system demonstrated strong capability in identifying clear-cut instances of pneumonia (achieving up to **99.9%** prediction confidence on positive test samples).

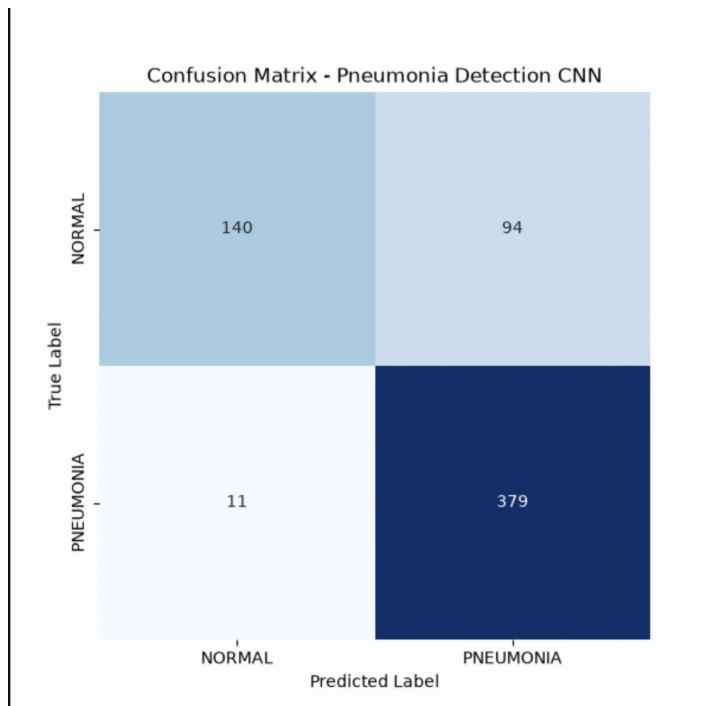
- **Model Behavior & Threshold Analysis:** While the model successfully recognizes characteristic opacities associated with pneumonia, borderline normal cases introduced slight false-positive sensitivities. Implementing a refined decision thresholding logic helped stabilize output certainty and minimize misclassifications on healthy lung scans.
- **Artifact Management:** The workflow effectively addresses data discrepancies, variable scanning conditions, and image scaling to maintain consistent tensor inputs.

5. Conclusion

In this project, we executed a comprehensive deep learning pipeline to analyze and classify chest X-ray images for pneumonia detection. Despite the challenges inherent in custom CNN training from scratch, the system successfully extracts critical visual features to predict patient status with high precision on clear cases.

The resulting framework provides a transparent, end-to-end blueprint—complete with automated inference, version-controlled architecture via GitHub, and reproducible notebooks—that can aid academic evaluation or serve as a foundational tool for automated medical screening research. Future enhancements could involve integrating advanced **Transfer Learning** architectures (such as MobileNetV2 or ResNet50) to further elevate generalization performance on borderline clinical scans.

Evaluation Results (Confusion Matrix)



Reference / Data Source

- Kaggle Chest X-Ray Images (Pneumonia) Dataset, sourced from pediatric and adult patient radiographic reports.